# ALI ISMAIL

**Software Engineer Intern | Agentic AI, LLM Evaluation and Observability on Google Cloud**

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**Portfolio: aliismail293.github.io** | *how-it-works walkthroughs and interactive system designs for every project*

## EXPERIENCE

**AI Engineering Intern** | TELUS Health, AI Accelerator | Toronto, ON — May 2026 – Aug 2026

- Shipped an LLM observability platform on self-hosted Langfuse covering health, usage, cost, quality and ownership for **24 production AI applications**, **replacing six separate systems** for **~30 users** across engineering, SRE and leadership.
- Built the team's first alerting on Cloudflare Workers, firing only on state transitions with severity routing and recovery notices; **13 of 33 tracked projects** sat in constant breach, so per-breach alerts would have fired daily.
- Designed a shared health engine for answer quality, hallucination rate and P95 latency with per-project overrides, read by the UI, alerting and assistant; flagged **4 projects** below the quality threshold and **2** above the 10-second P95 limit.
- Built per-project LLM cost attribution mapping opaque Apigee project IDs to business units, plus a value-vs-cost view of active users against spend that informed **decommissioning 2 low-usage projects**.
- Built a grounded AI assistant over live telemetry using query rewriting, logical routing across four sources and metadata filtering rather than vector search, so answers carry current figures; used by **~65% of platform users**.
- Brokered BigQuery, Langfuse and GCP access through one server-side service account, so **~30 users hold no credentials**.
- Prototyped an agentic code-testing tool with multi-step tool-calling loops; diagnosed an **~8,700-line input** that overflowed the context window and caused hallucination, then designed a sub-agent running-summary memory strategy to fix it.

## PROJECTS

*Full walkthroughs and system designs: aliismail293.github.io*

**Waypoint** | Multi-agent degree-planning assistant on Google Cloud | Live on Cloud Run — Oct 2026

*Google ADK, Gemini (Vertex AI), Cloud Run, Cloud SQL (pgvector), BigQuery, Firestore, MCP, OpenTelemetry, Terraform*

- Designed and deployed a **multi-agent system on Google ADK**: an orchestrator delegates to policy, course-data, timetable and ReAct planner agents, and a self-critique agent raised valid degree plans from **2/6 to 4/6**.
- Built **hybrid RAG** (pgvector HNSW + Postgres full-text, reciprocal rank fusion) over calendar policy text and BigQuery course data, reaching **81% recall@5** and **67% answer accuracy** on a 21-question held-out set at **$0.011 per question**.
- Connected a legacy timetable source, BigQuery and Waypoint itself through **3 MCP integrations**, and gated every PR in GitHub Actions on accuracy, P95 latency and a prompt-injection and PII **safety suite (9/9 passing)**.
- Traced every agent and tool with **OpenTelemetry to Langfuse and Cloud Trace**; measured a memory optimisation that **raised cost 31%** and kept it out of production; extracted **fde-kit** and reused it for a second live app.

**ragaudit** | Open-source RAG evaluation and tuning tool | aliismail293.github.io/rag-tester — Sep 2026 – Oct 2026

*Python, Pydantic, SQLite, Typer, Jinja2, httpx, pytest, Docker, Ollama*

- Built and shipped an open-source RAG auditor that exposed **100% answer correctness against a 30.8% retrieval hit rate**: **9 of 13 passes were lucky** answers from model memory, confirmed by disabling retrieval.
- Added verified test-set generation (second-model checks, **0.5 lexical-overlap cap**) and a tuning loop scored once on a **code-locked test set** split by chunk; **272 tests** run with zero network calls.

**Who's Real?** | Multiplayer social-deduction game on Roblox | aliismail293.github.io — Oct 2026 – Present

*Luau (strict), Rojo, ProfileStore, StyLua, Selene, Roblox Studio*

- Built a server-authoritative game on a **12-service state machine** where clients never receive hidden roles; **31 unit tests** on pure rule modules and a **60 fps server with 15 bots**; launching Dec 2026.

## EDUCATION

**University of Toronto** | Bachelor's Degree — Expected 2029

Minors in Computer Science and Mathematical Sciences.

**Coursework 2026–27:** Software Design, Theory of Computation, Data Structures and Analysis, Linear Algebra, Multivariable Calculus, Probability and Statistics.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Java, JavaScript, Rust, Luau

**AI and LLM:** Multi-agent systems (Google ADK; ReAct, self-reflection, hierarchical delegation), MCP servers, hybrid RAG (pgvector, full-text, RRF), query rewriting, routing, LLM-as-a-judge, Gemini on Vertex AI, OpenAI API, Ollama

**LLMOps and Evaluation:** Langfuse, OpenTelemetry, Cloud Trace, cost-per-request and tokens/sec tracking, P95 monitoring, alerting, held-out eval sets, CI eval gates, safety testing

**Cloud and Data:** Google Cloud (Cloud Run, Cloud SQL, BigQuery, Firestore, Vertex AI, Secret Manager, IAM), Terraform, GitHub Actions, Docker, Cloudflare Workers and D1, Apigee, ETL, Git, Linux